

Does Fine Dust Reduce Seoul's Ttareungyi Rentals?

Group 20

[RESEARCH_DESIGN.SPEC]

We tested one question with one main hypothesis and three follow-up tests

CORE TEST

Null vs. alternative

H0 posits higher daily PM2.5 does not reduce rentals; H1 tests whether rentals fall significantly (one-sided test).

SUBGROUPS

Discretionary trips

Pre-specified follow-up: leisure riding—measured by Han River station share and round trips—responds more strongly on days off.

ESTIMATION

DML & robust errors

Log-OLS, multiple regression, and Double Machine Learning (DML) with 5-fold cross-fitting and cross-fitted standard errors.

[DATA_INTAKE.LOG]

727 days of rentals, air and weather

DATASET_SOURCES.CSV

Domain	Coverage & Pipeline	Obs Count
Rentals	80,803,663 rentals across 2,904 stations from Seoul Open Data Plaza (2024–2025)	727 days
Air quality	Citywide daily mean PM2.5 monitoring; 58 days classified as "Bad" (> 35 μg/m³)	58 bad days
Weather	KMA ASOS station 108: mean temp, daily precipitation, snow cover, wind speed	4 weather vars

AGGREGATED DAILY PANEL · 2024.01.01 TO 2025.12.31 · SEOUL OPEN DATA PLAZA & KMA STATION 108

[PROJECT WEB]

We turned the analysis into a website: Clear the Haze

A data-story page where visitors **predict**, **explore** and **test** the evidence themselves.

The complex DML analysis and three-dimensional daily data were translated into an interactive narrative to communicate the findings clearly to both policy makers and the public.

LIVE WEBSITE LINK

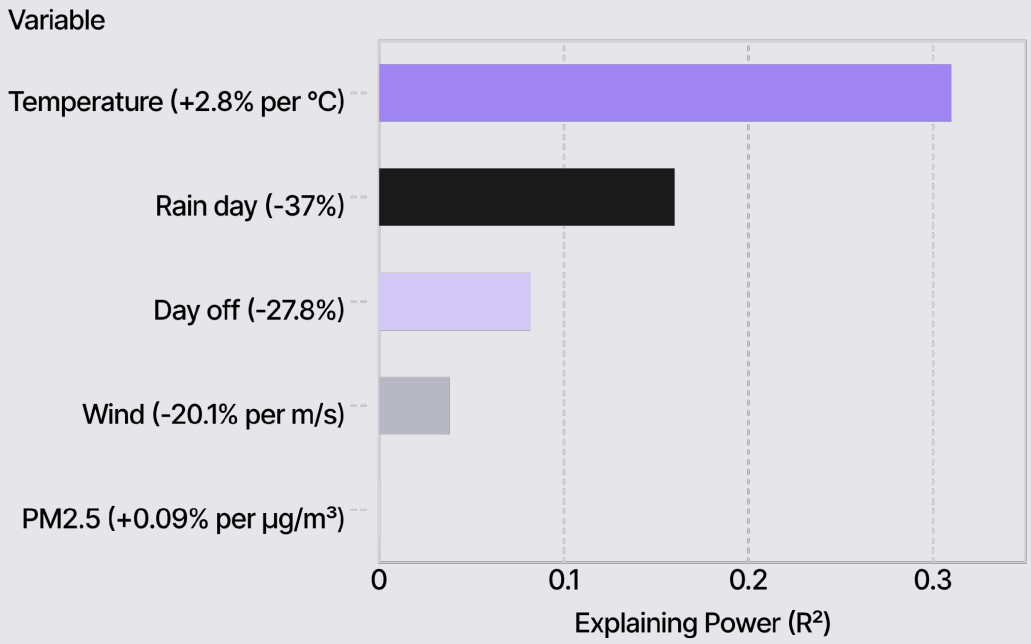
siwoo9573.github.io/ACDT-Group20-PM25-Ttareungyi



[UNIVARIATE_ANALYSIS.PLT]

One variable at a time, weather drives rentals and PM2.5 doesn't

BIVARIATE_R2_COMPARISON.PLT



MAP A FINDING

Temperature explains 31% of rental variation; PM2.5 explains 0.04%.

Snow: -13.9% per cm. Univariate slope for PM2.5 is slightly positive (+0.09% per µg/m³) with p = 0.69, showing no raw dampening signal whatsoever.

[CONFOUNDER_CHECK.PLT]

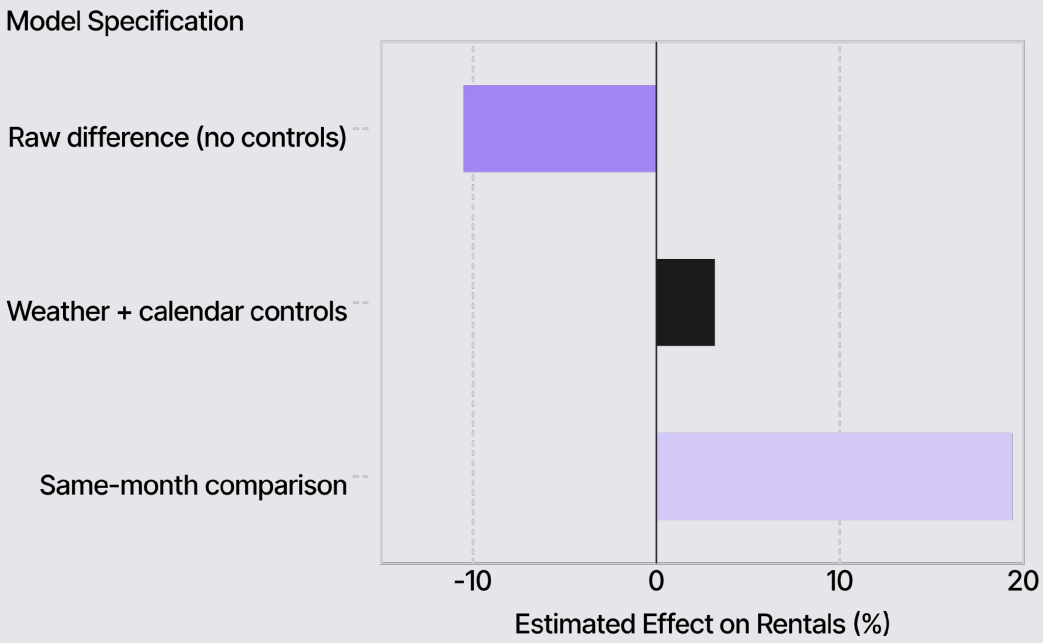
The raw gap looks real, until we add controls

REGRESSION SHIFT

Controlling for weather flips a -10.5% gap to +3.2%.

The apparent raw penalty is largely explained by when dusty days occur. Once temperature, precipitation, and month are held equal, the gap vanishes (p = 0.168; H0 not rejected).

BAD_DAY_ELASTICITY.PLT



[MECHANISM_DIAGNOSTICS]

Bad air falls on cold days

Season and weather act as powerful confounders, creating a negative gap that dust alone does not explain

88%

51 OF 58 BAD AIR DAYS FELL IN WINTER OR SPRING

7.8°C

AVG TEMP ON BAD DAYS VS. 16.1°C ON GOOD DAYS

-50%

WINTER BASELINE RENTALS COMPARED TO OTHER SEASONS

CONFOUNDING.TXT

The Omitted Bias

Winter conditions bring thermal inversions that trap PM2.5 over Seoul, while sub-zero weather simultaneously depresses cycling demand.

[PATH_VALIDATION.AUDIT]

Six what-if paths, only the confounder held up

HYPOTHESIS_FALSIFICATION_MATRIX.CSV

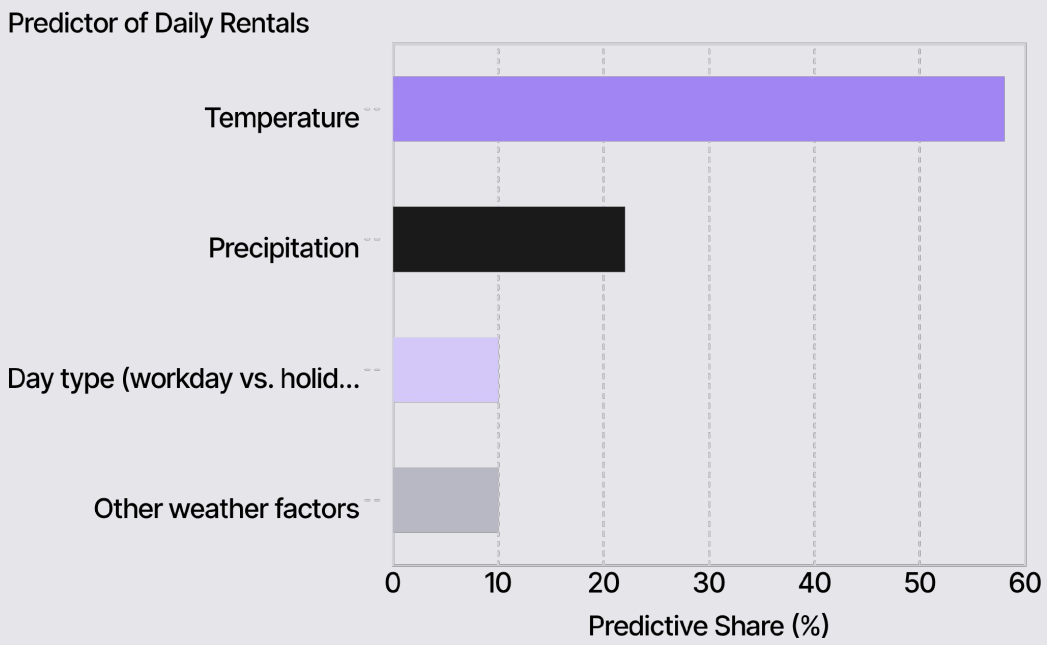
Path	Evidence	Evaluation
Confounder	Season and weather explain 42% of PM2.5 and 85% of rentals	**Supported**
Threshold above 35	p = 0.69	Not supported
Rain	p = 0.95	Not supported
Day off	p = 0.14–0.33	Not supported
Leisure use	p = 0.53–0.79	Not supported
Previous-day PM2.5	DML −0.6% to −0.8%, p ≥ 0.36	Not supported

NOTE: ALERTS (7 PM10 ALERT DAYS) AND COMMUTE-HOUR PEAKS WERE NOT TESTABLE.

[DML_ADJUSTED_ESTIMATE.PLT]

With drivers removed, the PM2.5 effect is near zero

FEATURE_IMPORTANCE_VS_EFFECT.PLT



DML RESULT

PER +10 MG/M³ PM2.5

-0.32%

95% CI [-1.83%, +1.21%]

DML finds no detectable city-wide effect, with the 95% CI bounded within roughly ±2%.

[EXPLORATORY_FINDING]

One lead: fewer Han River rides on bad workdays

HAN RIVER STATION SHARE	-1.17% per +10 μg/m³ (CI -1.75 to -0.58), survives Bonferroni correction
CAVEAT	Exploratory, not pre-specified; relative share only, absolute rentals -0.8%, p = 0.43

HAN_RIVER_CORRIDOR.IMG



[CONCLUSIONS_AND_LIMITS]

No detected effect is not proof of no effect

A non-significant result means an effect was not detected in daily aggregate data, bounded by three core constraints.

SAMPLE SIZE	Days off yielded -0.90% (CI [-2.70%, +0.93%]) with only 18 Bad days; statistical power remains too low to rule out leisure drops.
EXTREME HAZE	Only 13 days exceeded 50 μg/m³. Hourly peak commute dips could also be offset by shifts later in the day.
TRIP DURATION	Bad days show slightly shorter workday trips: 19.5 down to 19.1 minutes.

GITHUB: SIWOO9573/ACDT-GROUP20-PM25-TTAREUNGYI · WEB: SIWOO9573.GITHUB.IO/ACDT-GROUP20-PM25-TTAREUNGYI

STATION_TERMINAL.IMG



[RECOMMENDATIONS]

Plan bike supply on weather, not dust readings

Strategic Direction

- **Avoid Raw Data Bias:** Don't cite the raw -10.5% Bad-air day gap as proof of pollution damage; the gap is seasonal.
- **Prioritize Weather:** Temperature, rain and calendar account for most of the model's predictive importance (model R^2 85%). PM2.5 adds little prediction value.
- **Targeted Monitoring:** Watch Han River stations specifically on high-PM2.5 workdays; relative leisure share may dip.
- **Policy Evaluation:** Base city government interventions on weather- and season-adjusted estimates for accuracy.

WHAT-IF SCENARIO: MODAL SHIFT

If bad air actually pushed 2% of bike-km lost, 30% of it to cars, the environmental cost would be measurable.

DISTANCE LOST	5,200 km
CAR REPLACEMENT	30%
EXTRA EMISSIONS	~330 $kgCO_2eq$

**Based on Ministry of Environment (2024) car emission factor of 0.2111 kg CO_2eq per passenger-km.*

[STORY ARC]

Visitors guess first, then watch the gap disappear

The website leads users through an educational journey: starting with a **prediction**, revealing the raw **10.5% drop**, then methodically clearing the seasonal haze to find the adjusted effect.

STEP 01

Raw Comparison

−10.5%

Simple gap between Bad-air and Good-air days before any adjustments.

STEP 02

Same Month Only

+19.4%

Sign flips. The gap was an artifact of which months the bad air occurred in.

STEP 03

Weather Controls

+3.2%

Adding temp, rain, and snow makes the interval include zero. No clear effect.

STEP 04

Machine Learning

−0.32% per +10 µg/m³ PM2.5

95% CI [−1.83%, +1.21%]

Double ML puts the 95% CI within about ±2%. PM2.5 does not appear to be a major city-wide driver.

VISUAL HIGHLIGHT: "TURN THE DATA"

The page features a rotatable 3D scatter plot of 727 days. By filtering for workdays or days off, visitors see how temperature lines everything up while PM2.5 says almost nothing.

[REPRODUCIBILITY]

Open data and code, so anyone can reproduce it

GitHub Repository

```
# Install dependencies
pip install pandas numpy scikit-learn statsmodels matplotlib openpyxl

# Repository structure
/data # Merged daily data (727 days)
/code # DML notebook & round-trip script
/results # Figures & DML results

# Run analysis
jupyter nbconvert --execute Group20_DML_notebook.ipynb
```

All research logic, Han River station classifications, and DML cross-fitting seeds are public and version-controlled.

Data Sources

01 **SEOUL OPEN DATA PLAZA**
Ttareungyi daily rentals (OA-15246) and PM2.5 levels (OA-2220).

02 **KMA ASOS 108**
Daily temperature, humidity, rain, wind and snow cover in Seoul.

03 **OPENSTREETMAP (ODbL)**
Han River water polygons used for station proximity mapping.

04 **SEOUL STATION SNAPSHOTS**
Station coordinates (OA-13252).

REPOSITORY URL:
github.com/siwoo9573/ACDT-Group20-PM25-Ttareungyi

[EVIDENCE MATRIX]

Test a myth, check the leisure signal, see all the evidence

The Leisure Test

HAN RIVER CORRIDOR

Stations within 500m of the water show a **-1.2% dip** in relative share on bad workdays.

ROUND TRIP PATTERN

Same-station returns (leisure indicator) show **no significant response** to PM2.5, even on days off.

MYTH VS. REALITY

"Myth: Bad air pushes Seoul riders into cars."

Data: No detected rental loss to convert; the 10% raw gap is a seasonal pattern, not a pollution deterrent.

Full Evidence Table

TEST	QUESTION	VERDICT
Confounder	Do weather/season explain both air and rentals?	SUPPORTED
Threshold	Is there a sharp drop above 35 µg/m³?	NOT SUPPORTED
Day Off	Do people skip fun rides on bad-air weekends?	NOT SUPPORTED
Leisure Use	Do round trips decrease with dust?	NOT SUPPORTED
Lag Effect	Does yesterday's dust affect today's riding?	NOT SUPPORTED

**Percent effects are per +10 µg/m³ PM2.5. Caveat: Exploratory results (Han River share) are relative, not absolute reductions.*

FINAL ADJUSTED ESTIMATE

Fine dust isn't what keeps Seoul off its bikes

Effect per +10 µg/m³ PM2.5

-0.32%

95% CI [-1.83% to +1.21%]

STUDY LIMITS & NEXT STEPS

- **Sample size:** Only 58 Bad days in two years; power is low to rule out small weekend effects.
- **Aggregation:** Daily city totals may mask hourly peak commute-hour reactions.
- **Next:** Incorporating hourly rentals, alert timing, and transit card data for deeper insight.

INTERACTIVE WEBSITE

siwoo9573.github.io/...

OPEN SOURCE GITHUB

github.com/siwoo9573/...